

02_PCA — Principal Component Analysis

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1. What this notebook does

PCA serves two roles in this project (ISLR §12.2 / §12.4.3):

1. **Visualisation** — project customers onto PC1–PC2 to eyeball cluster structure later.
2. **Preprocessing** — replace ~7 correlated input features with 2–3 orthogonal PC scores to feed into k-means / hclust in 03_clustering.Rmd.

Output: ../data/customer_pc_scores.csv containing CustomerID + PC1..PCk (k = 3 by default).

2. Load and validate

```
customer <- read_csv("../data/processed/customer_features.csv",
                     show_col_types = FALSE)

stopifnot(
  "CustomerID" %in% names(customer),
  !anyDuplicated(customer$CustomerID),
  nrow(customer) > 0
)

glimpse(customer)
```

```
## Rows: 4,334
## Columns: 29
## $ CustomerID      <dbl> 12346, 12347, 12348, 12349, 12350, 12352, 12353, 1~
## $ Recency          <dbl> 326, 2, 75, 19, 310, 36, 204, 232, 214, 23, 33, 2,~
## $ Frequency        <dbl> 1, 7, 4, 1, 1, 7, 1, 1, 1, 3, 1, 2, 4, 3, 1, 10, 2~
## $ GrossMonetary    <dbl> 77183.60, 4310.00, 1437.24, 1457.55, 294.40, 1385.~
## $ TotalQuantity    <dbl> 74215, 2458, 2332, 630, 196, 526, 20, 530, 240, 15~
## $ UniqueProducts   <dbl> 1, 103, 21, 72, 16, 57, 4, 58, 13, 52, 131, 12, 21~
## $ FirstPurchase    <dtm> 2011-01-18 10:01:00, 2010-12-07 14:57:00, 2010-12~
## $ LastPurchase     <dtm> 2011-01-18 10:01:00, 2011-12-07 15:52:00, 2011-09~
## $ TenureDays       <dbl> 0, 365, 282, 0, 0, 260, 0, 0, 0, 302, 0, 149, 274,~
## $ SinglePurchase    <dbl> 1, 0, 0, 1, 1, 0, 1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0,~
## $ AvgUnitPrice      <dbl> 1.0400000, 1.7534581, 0.6163122, 2.3135714, 1.5020~
## $ AvgOrderValue     <dbl> 77183.6000, 615.7143, 359.3100, 1457.5500, 294.400~
```

```
## $ AvgOrderItems      <dbl> 74215.00000, 351.14286, 583.00000, 630.00000, 196.~
## $ ReturnInvoices     <dbl> 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 2, 0, 0, 3, 0,~
## $ ReturnValueAbs     <dbl> 77183.60, 0.00, 0.00, 0.00, 0.00, 120.33, 0.00, 0.~
## $ Country            <chr> "United Kingdom", "Iceland", "Finland", "Italy", "~
## $ NetMonetary        <dbl> 0.00, 4310.00, 1437.24, 1457.55, 294.40, 1265.41, ~
## $ ReturnRate         <dbl> 0.5000000, 0.0000000, 0.0000000, 0.0000000, 0.0000~
## $ ReturnValueRate    <dbl> 1.000000000, 0.000000000, 0.000000000, 0.000000000~
## $ AvgUnitPrice_winsor <dbl> 1.0400000, 1.7534581, 0.6163122, 2.3135714, 1.5020~
## $ log_Recency        <dbl> 5.789960, 1.098612, 4.330733, 2.995732, 5.739793, ~
## $ log_Frequency      <dbl> 0.6931472, 2.0794415, 1.6094379, 0.6931472, 0.6931~
## $ log_GrossMonetary  <dbl> 11.253955, 8.368925, 7.271175, 7.285198, 5.688330,~
## $ log_TotalQuantity  <dbl> 11.214735, 7.807510, 7.754910, 6.447306, 5.283204,~
## $ log_UniqueProducts <dbl> 0.6931472, 4.6443909, 3.0910425, 4.2904594, 2.8332~
## $ log_AvgOrderValue  <dbl> 11.253955, 6.424406, 5.886965, 7.285198, 5.688330,~
## $ log_NetMonetary    <dbl> 0.000000, 8.368925, 7.271175, 7.285198, 5.688330, ~
## $ Monetary           <dbl> 77183.60, 4310.00, 1437.24, 1457.55, 294.40, 1385.~
## $ log_Monetary       <dbl> 11.253955, 8.368925, 7.271175, 7.285198, 5.688330,~
```

3. Modelling matrix

The seven-feature input drops everything that the EDA flagged as redundant or distortive:

- `log_NetMonetary` instead of `log_GrossMonetary` — captures the customer's actual contribution to revenue net of returns.
- `AvgUnitPrice_winsor` instead of raw `AvgUnitPrice` — without winsorising, a few extreme items dominate PC2/PC3 loadings.
- `AvgOrderValue` kept (drop `AvgBasketValue`, which was an inconsistent line-level proxy in earlier iterations and is no longer produced by `dataPreprocess.py`).
- `Country` excluded — UK dominance would force a near-degenerate split.
- `ReturnRate` / `ReturnValueRate` held back as profiling variables (sparse, mostly zero).

```
feature_cols <- c("log_Recency", "log_Frequency", "log_NetMonetary",
                  "log_UniqueProducts", "log_AvgOrderValue",
                  "AvgUnitPrice_winsor", "TenureDays")

X <- customer |>
  select(CustomerID, all_of(feature_cols)) |>
  drop_na()

# Sanity-check: row count must equal customer count.
stopifnot(nrow(X) == nrow(customer))

# Confirm winsorisation worked - pre-winsor max can be hundreds of pounds.
summary(X$AvgUnitPrice_winsor)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.08562 1.39329 1.78295 2.02562 2.31495 7.60380
```

```
quantile(X$AvgUnitPrice_winsor, c(.5, .9, .99, 1))
```

```
##      50%      90%      99%     100%
## 1.782945 3.210570 7.572846 7.603800
```

```
dim(X)
```

```
## [1] 4334      8
```

```
summary(X[, feature_cols])
```

```
##   log_Recency    log_Frequency    log_NetMonetary    log_UniqueProducts
##   Min.   :0.6931    Min.   :0.6931    Min.   : -4.959    Min.   :0.6931
##   1st Qu.:2.9444    1st Qu.:0.6931    1st Qu.: 5.696    1st Qu.:2.8332
##   Median :3.9512    Median :1.0986    Median : 6.474    Median :3.5835
##   Mean   :3.8321    Mean   :1.3416    Mean   : 6.532    Mean   :3.5496
##   3rd Qu.:4.9698    3rd Qu.:1.7918    3rd Qu.: 7.379    3rd Qu.:4.3567
##   Max.   :5.9269    Max.   :5.3327    Max.   :12.538    Max.   :7.4877
##   log_AvgOrderValue AvgUnitPrice_winsor    TenureDays
##   Min.   : 1.558    Min.   :0.08562    Min.   : 0.0
##   1st Qu.: 5.183    1st Qu.:1.39329    1st Qu.: 0.0
##   Median : 5.672    Median :1.78295    Median : 92.0
##   Mean   : 5.638    Mean   :2.02562    Mean   :130.3
##   3rd Qu.: 6.051    3rd Qu.:2.31495    3rd Qu.:252.0
##   Max.   :11.341    Max.   :7.60380    Max.   :373.0
```

4. Fit PCA

`prcomp(center = TRUE, scale. = TRUE)` standardises every column to zero mean / unit variance — required because `AvgUnitPrice_winsor` and `TenureDays` are on totally different scales from the log features.

```
pr.out <- prcomp(X[, feature_cols], center = TRUE, scale. = TRUE)
pr.out
```

```
## Standard deviations (1, .., p=7):
## [1] 1.9208932 1.0595938 1.0043846 0.7323402 0.6042268 0.4637281 0.2493717
##
## Rotation (n x k) = (7 x 7):
##           PC1          PC2          PC3          PC4          PC5
## log_Recency   -0.3543127 -0.3247000 -0.11967389 -0.86652203 -0.05794226
## log_Frequency   0.4588637  0.2848157 -0.08419794 -0.25300531 -0.16465563
## log_NetMonetary  0.4729638 -0.2306870 -0.18640031 -0.07994676 -0.22244905
## log_UniqueProducts 0.4243305 -0.1557292  0.14121516 -0.19256758  0.85444813
## log_AvgOrderValue 0.2472991 -0.7649580 -0.22875620  0.23764198 -0.19657842
## AvgUnitPrice_winsor -0.1137232  0.2058930 -0.93179092  0.08002433  0.26238229
## TenureDays      0.4313532  0.3292859 -0.05777853 -0.28067156 -0.28719068
##           PC6          PC7
## log_Recency   0.008231418 -0.020618134
## log_Frequency   0.528447887 -0.575199481
## log_NetMonetary  0.352382528  0.712948214
## log_UniqueProducts -0.092567368 -0.004207856
## log_AvgOrderValue -0.219253920 -0.397698340
## AvgUnitPrice_winsor -0.033990784  0.006087152
## TenureDays     -0.733965432  0.046974079
```

5. Variance explained

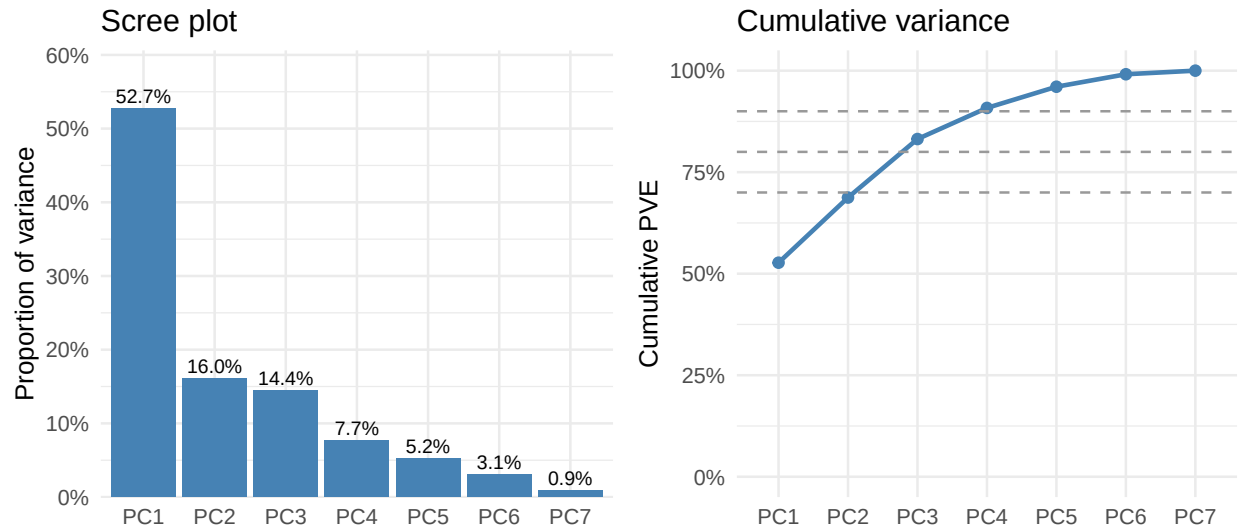
```
pve <- pr.out$sdev^2 / sum(pr.out$sdev^2)
pve_df <- tibble(
  PC      = factor(paste0("PC", seq_along(pve)),
    levels = paste0("PC", seq_along(pve))),
  PVE      = pve,
  cumulative = cumsum(pve)
)
pve_df
```

```
## # A tibble: 7 x 3
##   PC      PVE cumulative
##   <fct>   <dbl>     <dbl>
## 1 PC1    0.527      0.527
## 2 PC2    0.160      0.688
## 3 PC3    0.144      0.832
## 4 PC4    0.0766     0.908
## 5 PC5    0.0522     0.960
## 6 PC6    0.0307     0.991
## 7 PC7    0.00888    1
```

```
p_screes <- ggplot(pve_df, aes(PC, PVE)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = percent(PVE, accuracy = 0.1)),
    vjust = -0.4, size = 3) +
  scale_y_continuous(labels = percent_format(),
    expand = expansion(mult = c(0, 0.15))) +
  labs(title = "Scree plot", x = NULL, y = "Proportion of variance")

p_cum <- ggplot(pve_df, aes(PC, cumulative, group = 1)) +
  geom_line(colour = "steelblue", linewidth = 1) +
  geom_point(colour = "steelblue", size = 2) +
  geom_hline(yintercept = c(0.7, 0.8, 0.9),
    linetype = "dashed", colour = "grey60") +
  scale_y_continuous(labels = percent_format(), limits = c(0, 1)) +
  labs(title = "Cumulative variance", x = NULL, y = "Cumulative PVE")

gridExtra::grid.arrange(p_screes, p_cum, nrow = 1)
```



```
tibble(
  PC      = paste0("PC", seq_along(pr.out$sdev)),
  eigenvalue = pr.out$sdev^2,
  keep_kaiser = pr.out$sdev^2 > 1
)
```

```
## # A tibble: 7 x 3
##   PC      eigenvalue keep_kaiser
##   <chr>      <dbl> <lgl>
## 1 PC1        3.69  TRUE
## 2 PC2        1.12  TRUE
## 3 PC3        1.01  TRUE
## 4 PC4        0.536 FALSE
## 5 PC5        0.365 FALSE
## 6 PC6        0.215 FALSE
## 7 PC7        0.0622 FALSE
```

Kaiser's rule (eigenvalue > 1) gives a quick orthogonal check on the elbow / variance-threshold heuristic. We typically keep the first three PCs across all three rules.

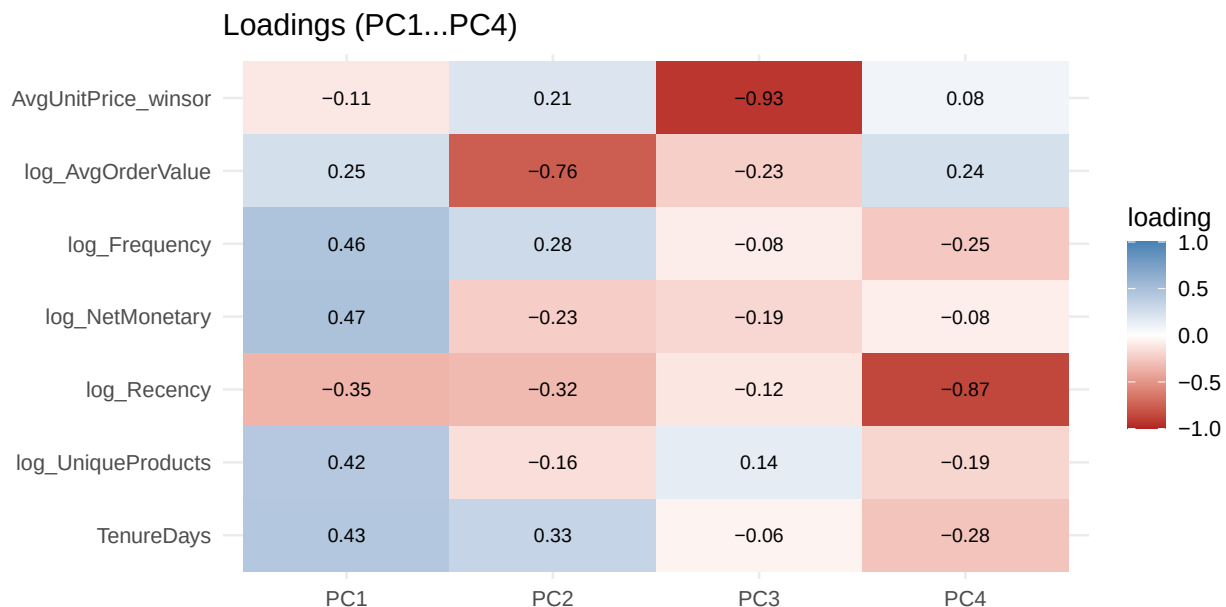
6. Loadings — what each PC actually means

```
loadings <- as_tibble(pr.out$rotation, rownames = "feature")
loadings |> mutate(across(where(is.numeric), \ (x) round(x, 2)))
```

```
## # A tibble: 7 x 8
##   feature      PC1  PC2  PC3  PC4  PC5  PC6  PC7
##   <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 log_Recency -0.35 -0.32 -0.12 -0.87 -0.06  0.01 -0.02
## 2 log_Frequency  0.46  0.28 -0.08 -0.25 -0.16  0.53 -0.58
## 3 log_NetMonetary  0.47 -0.23 -0.19 -0.08 -0.22  0.35  0.71
## 4 log_UniqueProducts  0.42 -0.16  0.14 -0.19  0.85 -0.09  0
```

```
## 5 log_AvgOrderValue      0.25 -0.76 -0.23  0.24 -0.2  -0.22 -0.4
## 6 AvgUnitPrice_winsor -0.11  0.21 -0.93  0.08  0.26 -0.03  0.01
## 7 TenureDays             0.43  0.33 -0.06 -0.28 -0.29 -0.73  0.05
```

```
loadings |>
  pivot_longer(-feature, names_to = "PC", values_to = "loading") |>
  filter(PC %in% paste0("PC", 1:4)) |>
  mutate(PC = factor(PC, levels = paste0("PC", 1:4))) |>
  ggplot(aes(PC, fct_rev(feature), fill = loading)) +
  geom_tile() +
  geom_text(aes(label = sprintf("%.2f", loading)), size = 3) +
  scale_fill_gradient2(low = "firebrick", mid = "white",
    high = "steelblue", midpoint = 0,
    limits = c(-1, 1)) +
  labs(title = "Loadings (PC1-PC4)", x = NULL, y = NULL)
```



Note on signs. PCA component signs are arbitrary — `prcomp` could have flipped any column. What matters is the *contrast* across features, not the absolute sign. Read each PC as “high vs. low” along the loadings displayed below.

Reading the loadings on the cleaned/winsorised features:

- **PC1 — engagement / customer value.** Loads positively on `log_Frequency`, `log_NetMonetary`, `log_UniqueProducts`, and `TenureDays`, and negatively on `log_Recency`. High PC1 = recent, frequent, broad-product, repeat-relationship customers. Low PC1 = lapsed one-time buyers.
- **PC2 — order-size vs. tenure contrast.** Strong negative loading on `log_AvgOrderValue` and `AvgUnitPrice_winsor`, positive on `TenureDays` and `log_Frequency`. Low PC2 ~ wholesale-leaning behaviour (fewer, larger orders of higher-priced items). High PC2 ~ smaller, more frequent transactions over a longer relationship. **It is *not* purely a “tenure / lifecycle” axis** — describing it that way ignores the `AvgOrderValue` contrast, which is the more informative half.
- **PC3 — residual price-point.** Once PC1 and PC2 absorb engagement and order-size, PC3 picks up what’s left of `AvgUnitPrice_winsor` — essentially the price tier the customer typically buys at.

7. PC scores — the new feature space

```
scores <- as_tibble(pr.out$x) |>
  mutate(CustomerID = X$CustomerID, .before = 1) |>
  left_join(customer |> select(CustomerID, GrossMonetary, NetMonetary,
                             Frequency, Recency, Country,
                             ReturnRate, ReturnValueRate, SinglePurchase),
            by = "CustomerID")

stopifnot(nrow(scores) == nrow(X), !anyDuplicated(scores$CustomerID))
head(scores)
```

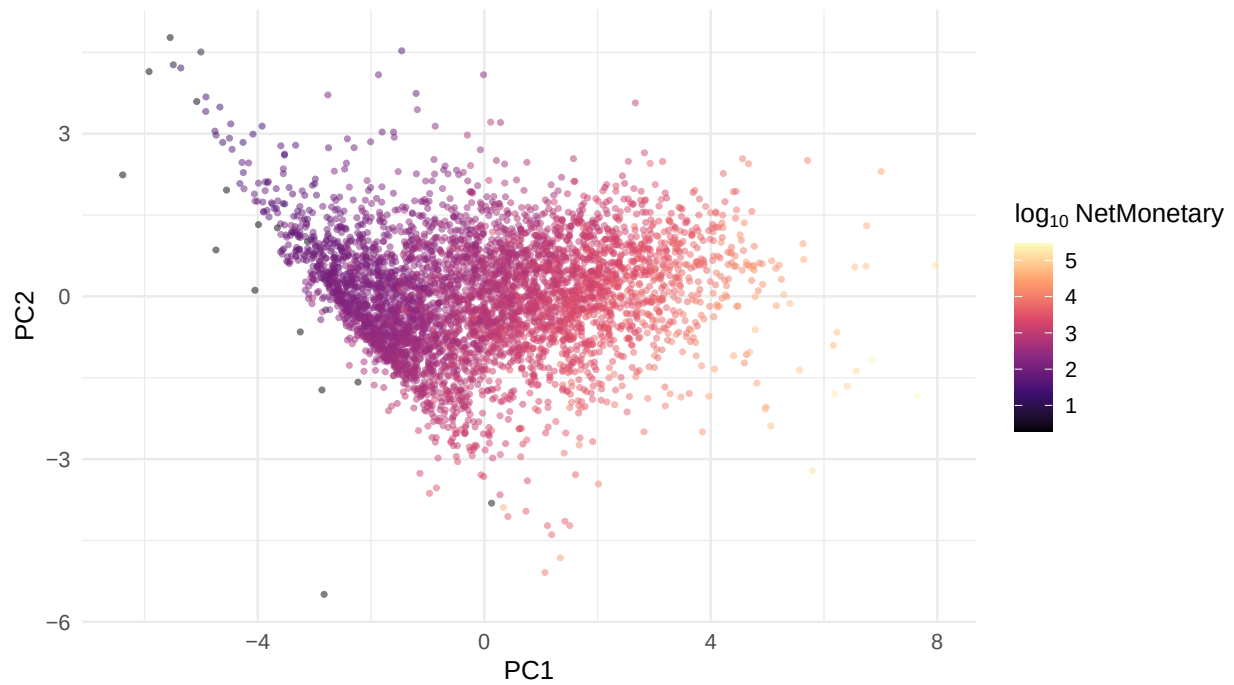
```
## # A tibble: 6 x 16
##   CustomerID  PC1    PC2    PC3    PC4    PC5    PC6    PC7 GrossMonetary
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 12346 -2.83 -5.49 -0.371 1.86 -2.41 -2.91 -6.06 77184.
## 2 12347 3.34 0.224 -0.0869 0.928 -0.327 -0.571 0.0716 4310
## 3 12348 0.864 -0.214 0.800 -0.733 -1.29 -0.427 0.0816 1437.
## 4 12349 0.426 -2.27 -0.551 1.43 0.540 -0.137 0.0393 1458.
## 5 12350 -1.87 -0.960 0.421 -0.564 -0.177 0.0671 -0.0123 294.
## 6 12352 1.21 0.976 -0.555 -0.597 0.0665 0.0549 -0.0555 1386.
## # i 7 more variables: NetMonetary <dbl>, Frequency <dbl>, Recency <dbl>,
## #   Country <chr>, ReturnRate <dbl>, ReturnValueRate <dbl>,
## #   SinglePurchase <dbl>
```

7a. PC1 vs PC2, coloured by net spend

```
ggplot(scores, aes(PC1, PC2,
                   colour = log10(pmax(NetMonetary, 1) + 1))) +
  geom_point(alpha = 0.5, size = 0.8) +
  scale_colour_viridis_c(option = "magma",
                        name = expression(log[10]-NetMonetary)) +
  labs(title = "Customers in PC1-PC2 space, coloured by net spend",
       subtitle = "PC1 should align with engagement / customer value")
```

Customers in PC1...PC2 space, coloured by net spend

PC1 should align with engagement / customer value

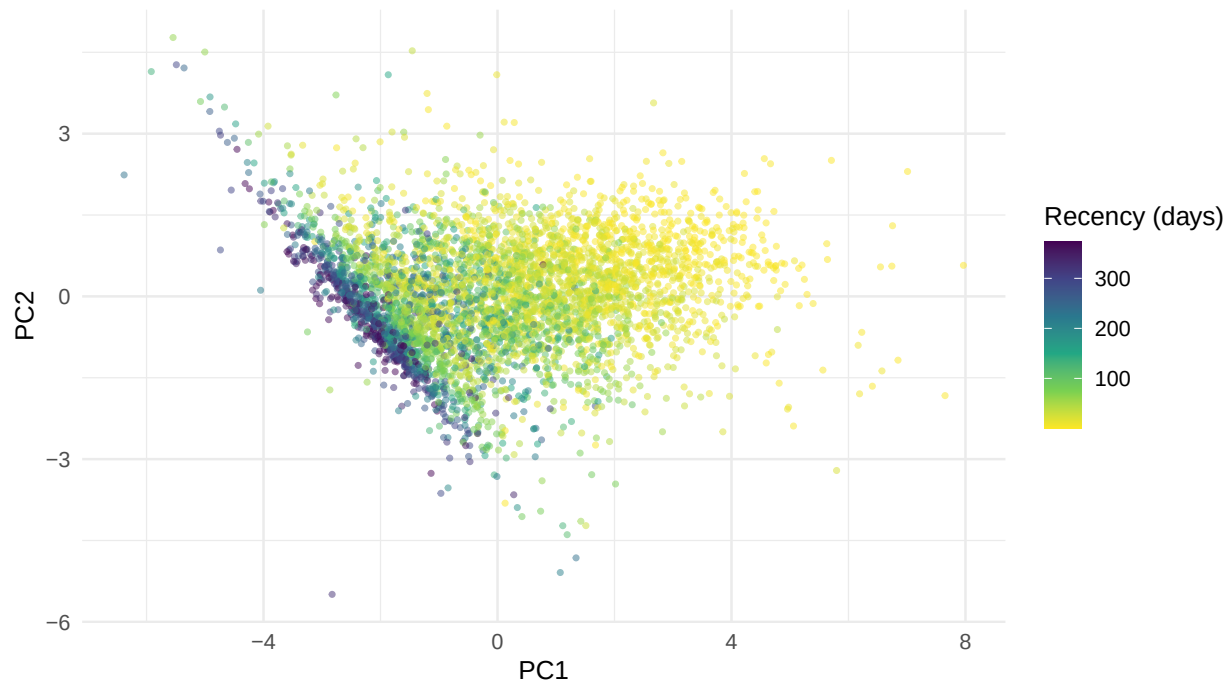


7b. PC1 vs PC2, coloured by recency

```
ggplot(scores, aes(PC1, PC2, colour = Recency)) +  
  geom_point(alpha = 0.5, size = 0.8) +  
  scale_colour_viridis_c(option = "viridis", direction = -1,  
    name = "Recency (days)") +  
  labs(title = "PC1-PC2 space, coloured by recency",  
    subtitle = "Recent customers should sit on the engagement side of PC1")
```


PC1...PC2 space, coloured by recency

Recent customers should sit on the engagement side of PC1



8. Biplot

A ggplot biplot reads more clearly than `biplot(pr.out)` for the report.

```
arrow_scale <- with(pr.out,
  min(max(abs(x[, 1])) / max(abs(rotation[, 1])),
    max(abs(x[, 2])) / max(abs(rotation[, 2])))) * 0.7

arrows_df <- as_tibble(pr.out$rotation, rownames = "feature") |>
  mutate(PC1 = PC1 * arrow_scale,
    PC2 = PC2 * arrow_scale)

ggplot(scores, aes(PC1, PC2)) +
  geom_point(alpha = 0.25, size = 0.6, colour = "grey50") +
  geom_segment(data = arrows_df,
    aes(x = 0, y = 0, xend = PC1, yend = PC2),
    arrow = arrow(length = unit(0.2, "cm")),
    colour = "firebrick", linewidth = 0.7) +
  geom_text(data = arrows_df,
    aes(x = PC1 * 1.08, y = PC2 * 1.08, label = feature),
    colour = "firebrick", size = 3.5) +
  labs(title = "Biplot - customers + variable loadings on PC1-PC2",
    x = sprintf("PC1 (%s)", percent(pve[1], accuracy = 0.1)),
    y = sprintf("PC2 (%s)", percent(pve[2], accuracy = 0.1)))
```

Biplot ... customers + variable loadings on PC1...PC2

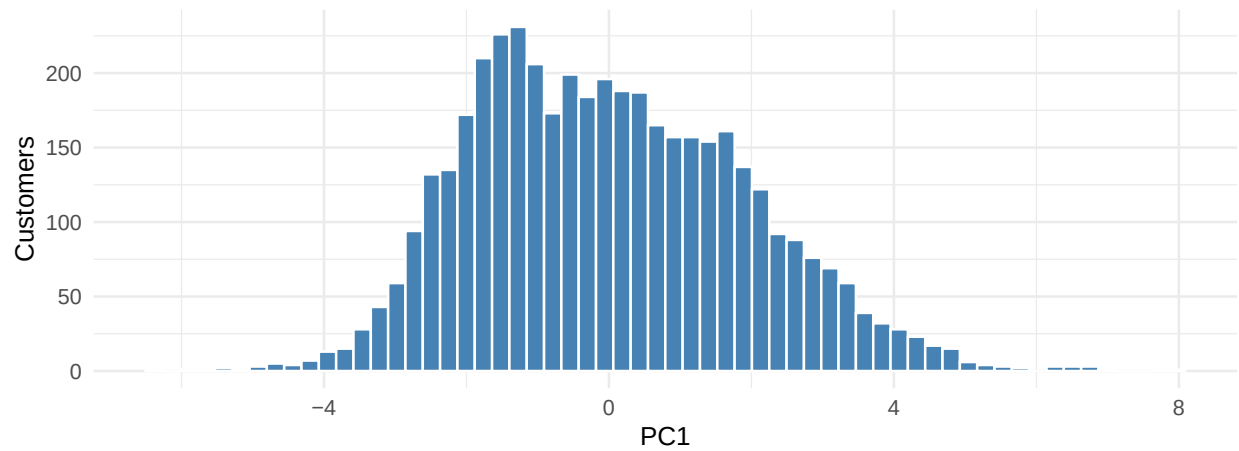


9. PC1 sanity check — symmetric core, real upper tail

```
ggplot(scores, aes(PC1)) +
  geom_histogram(bins = 60, fill = "steelblue", colour = "white") +
  labs(title = "Distribution of PC1 (engagement axis)",
       subtitle = paste("Approximately symmetric in the bulk,",
                        "but a real upper-shoulder tail of high-value customers"),
       x = "PC1", y = "Customers")
```

Distribution of PC1 (engagement axis)

Approximately symmetric in the bulk, but a real upper-shoulder tail of high-value customers



PC1 is approximately symmetric in the bulk but retains a right shoulder representing the wholesale-leaning tail. We do **not** claim the distribution is “well-behaved eno